

Auteur: Stan Van Campenhout

Studierichting: Informatica

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$$\begin{aligned} \int \frac{5 + \sqrt{2x+3}}{3 - \sqrt{2x+3}} dx &\stackrel{t=\sqrt{2x+3}}{=} \int \frac{5+t}{3-t} t dt = - \int \frac{t^2 + 5t}{t-3} dt \\ &= - \int \left(t + 8 + \frac{24}{t-3} \right) dt = -\frac{t^2}{2} - 8t - 24 \int \frac{1}{t-3} dt \\ &= -\frac{t^2}{2} - 8t - 24 \ln |t-3| + c \stackrel{t=\sqrt{2x+3}}{=} -x - 8\sqrt{2x+3} - 24 \ln |\sqrt{2x+3} - 3| + c \end{aligned}$$

References